

ON THE DETERMINATION OF THE HEIGHT OF THE EKMAN BOUNDARY LAYER

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Abstract. The height h_τ of the Ekman turbulent boundary layer determined by the momentum flux profile is estimated with the aid of considerations of similarity and an analysis of the dynamic equations. Asymptotic formulae have been obtained showing that, with increasing instability, h_τ increases as $|\mu|^{1/2}$ (where μ is the non-dimensional stratification parameter); with increasing stability, on the other hand, h decreases as $\mu^{-1/2}$. For comparison, a simple estimate of the boundary-layer height h_u determined by the velocity profile is given. As is shown, in unstable stratification, h_u behaves asymptotically as $|\mu|^{-1}$, i.e., in a manner entirely different from that of h_τ .

Аннотация. Толщина турбулентного экмановского слоя h_τ , определяемая по профилю потока импульса, оценивается с помощью соображений подобия и анализа динамических уравнений. Получены асимптотические формулы, согласно которым h_τ с усилением неустойчивости растет как $|\mu|^{1/2}$ (μ – безразмерный параметр стратификации), а с усилением устойчивости убывает как $\mu^{-1/2}$. Для сравнения приведена простая оценка толщины пограничного слоя h_u , определяемой по профилю скорости. Показано, что при неустойчивой стратификации h_u асимптотически ведет себя как $|\mu|^{-1}$, т.е., совершенно иначе, чем h_τ .

For a turbulent Ekman layer, i.e., a horizontally homogeneous and steady turbulent boundary layer in a stratified rotating fluid, the averaged equations of motion are as follows:

$$f(v - V_g) + \frac{1}{\rho} \frac{\partial \tau_x}{\partial z} = 0, \quad -f(u - U_g) + \frac{1}{\rho} \frac{\partial \tau_y}{\partial z} = 0, \quad (1)$$

where ρ is the density; f – the Coriolis parameter; z – the vertical coordinate calculated from the underlying surface; u and v – the mean velocity components along the x and y horizontal axes; U_g and V_g – the corresponding geostrophic velocities; τ_x and τ_y – the momentum flux components or Reynolds stresses determined by the formulae $\tau_x = -\rho \overline{u'w'}$, $\tau_y = -\rho \overline{v'w'}$, where u' , v' , and w' are turbulent velocity components along the x , y , and z axes, and the bar indicates averaging. The τ_x and τ_y values are often expressed by the Boussinesq formulae:

$$\tau_x = \rho k \frac{\partial u}{\partial z}, \quad \tau_y = \rho k \frac{\partial v}{\partial z}, \quad (2)$$

where k is an eddy viscosity. Consequently, boundary-layer research has been concerned chiefly with finding the dependence of k on mean flow characteristics. The mean velocity and momentum flux components satisfy the following boundary conditions:

$$u = v = 0 \quad \text{at} \quad z = z_0; \quad u \rightarrow U_g, \quad v \rightarrow V_g \quad \text{at} \quad z \rightarrow \infty, \quad (3)$$

where z_0 is the roughness parameter of the underlying surface.

To be precise we shall hereafter refer to the motion of a temperature-stratified atmosphere assuming that the x axis is directed along the surface momentum flux vector. The conditions of a steady and horizontally homogeneous flow promptly yield height independence of the vertical heat flux $H = c_p \rho \overline{T'w'}$ (ρ and c_p being respectively air density and thermal capacity at constant pressure, and T' being temperature fluctuation). Further, according to the similarity theory (Monin and Obukhov, 1954; Kazanski and Monin, 1960) the statistical characteristics of the velocity and temperature fields are determined by the following four parameters: *friction velocity* $u_* = \sqrt[4]{[(\tau_x^2 + \tau_y^2)/\rho^2]} = \sqrt{[\tau_{x0}/\rho]}$, *normalized heat flux* $Q = H/c_p \rho$, *buoyancy parameter* $\beta = g/T_0$ (g – gravity acceleration, T_0 – typical value of air temperature), and *Coriolis parameter* f . Two independent combinations of length dimension may be composed of the above parameters:

$$L = -\frac{u_*^3}{\kappa \beta Q}, \quad \lambda = \frac{\kappa u_*}{f}. \quad (4)$$

The first combination is the Monin–Obukhov length-scale and the second one is an additional scale associated with rotation ($\kappa \approx 0.4$ is the von Karman constant, traditionally introduced into expressions for L and λ). Thus, the turbulent regime under consideration proves to be dependent on a single dimensionless stratification parameter:

$$\mu = \frac{\lambda}{L}; \quad (5)$$

and the general expressions for the profiles of the horizontal velocity components and the components of the vertical flux of momentum may be shown to be:

$$\begin{aligned} u - U_g &= u_* \Psi_u \left(\frac{z}{\lambda}, \mu \right), & v - V_g &= u_* \Psi_v \left(\frac{z}{\lambda}, \mu \right), \\ \tau_x &= \rho u_*^2 \Psi_x \left(\frac{z}{\lambda}, \mu \right), & \tau_y &= \rho u_*^2 \Psi_y \left(\frac{z}{\lambda}, \mu \right), \end{aligned} \quad (6)$$

where Ψ_u , Ψ_v and Ψ_x , Ψ_y are dimensionless universal functions of the two arguments. The functions tend towards zero at $z/\lambda \rightarrow \infty$ in compliance with (3).

In the case of neutral stratification, Formulae (6) reveal full similarity with the velocity defect law for turbulent flow in a round pipe (see Monin and Yaglom, 1971, Chapter 3), the geostrophic velocity acting in the same manner as a pipe axial velocity and the λ scale acting as a pipe radius. With neutral stratification, therefore, the Ekman boundary-layer thickness proves to be proportional to λ (the same conclusion might be drawn in a more direct way: with neutral stratification, the turbulent conditions are determined by two parameters only – u_* and f , while λ is a single independent combination of the length dimensionality which can be composed from these parameters).

In the general case, the Ekman boundary-layer thickness h_u – for velocity and h_τ – for momentum flux, ought to be determined from the condition that the main variations of the vectors (u, v) and (τ_x, τ_y) in the semi-space $z_0 < z < \infty$ remain within the height ranges $z_0 < z < h_u$ and $z_0 < z < h_\tau$, respectively. In other words, assuming tolerable relative errors δ_u and δ_τ and using u_* as a velocity scale, we should obtain h_u and h_τ from the conditions

$$\begin{aligned} \sqrt{(u - U_g)^2 + (v - V_g)^2}/u_* &< \delta_u \quad \text{for } z > h_u, \\ \sqrt{(\tau_x^2 + \tau_y^2)}/\rho u_*^2 &< \delta_\tau \quad \text{for } z > h_\tau. \end{aligned} \quad (7)$$

Generally speaking, the Ekman boundary layer, in contrast with a turbulent boundary layer over a flat plate, has no well-defined upper surface and its thickness may be determined in many different ways. In the flow over a thin plate, two quantities are usually taken into consideration, namely, the displacement thickness h^* and the momentum loss thickness h^{**} , which are defined by the following expressions:

$$h^* = \int_0^\infty (1 - U/G) dz, \quad h^{**} = \int_0^\infty (U/G) (1 - U/G) dz,$$

where U is the velocity at a distance z from the plane, and G is the free-stream velocity. These definitions of h^* and h^{**} can be easily applied to the case of an Ekman boundary-layer using $U = \sqrt{(u^2 + v^2)}$ and $G = \sqrt{(U_g^2 + V_g^2)}$; alternatively, (see Hanna, 1969) h^* and h^{**} can be considered separately for the wind-speed components, one of which is parallel and the other is at right angles to the isobars. Finally, when determining h_u by means of the formula similar to the first one of (7), an absolute value of the geostrophic wind G can be used as a velocity scale instead of the friction velocity u_* . This seems more natural from the point of view of estimating the proximity of the real wind speed to the geostrophic one; but it is less convenient since in that case, h_u/λ depends not only on μ but also on the surface Rossby number $Ro = G/fz_0$. Further, we are going to consider only the definitions based on (7). Unfortunately, therefore, we cannot use the vast amount of experimental data analysed by Hanna (1969).

According to (6) and (7), ratios h_u/λ and h_τ/λ both depend only on μ , i.e.,

$$h_u = \lambda \varphi_u(\mu), \quad h_\tau = \lambda \varphi_\tau(\mu), \quad (8)$$

where φ_u and φ_τ are non-dimensional universal functions of one argument, the dimensional analysis being, of course, insufficient for defining their type. We shall attempt to define asymptotic behaviour of $\varphi_\tau(\mu)$ at large positive and negative μ 's using the dynamic equations.

The approach may be explained by analysing first a simple case: a laminar Ekman layer. The appropriate equations of motion may be obtained from (1) and (2), provided a kinematic viscosity ν is substituted for an eddy viscosity k .

Since ν is a known constant, the system (1)–(3) can easily be solved, the non-dimen-

sional momentum flux being expressed by the formula

$$\sqrt{\tau_x^2 + \tau_y^2} / \rho u_*^2 = \exp\left(-\frac{z}{\sqrt{2\nu/f}}\right). \quad (9)$$

So the height h_τ found by using the second part of formulae (7) is given by the expression* $h_\tau \sim \sqrt{\nu/f}$. (The numerical proportionality coefficient depends, naturally, on the assumed tolerable error δ_τ .) To estimate the height h_τ for a turbulent Ekman layer, a similar formula might be employed:

$$h_\tau \sim \sqrt{K/f}, \quad (10)$$

where K is an effective value of the eddy viscosity.

We shall commence with neutral stratification, i.e., $\mu=0$. In this instance, at least in the lower layers, $k = \kappa u_* z$; by analogy $K = \kappa u_* h_\tau$ may be assumed as an effective value of K . Substituting this expression in (10), we obtain a familiar result:

$$h_\tau \sim \frac{\kappa u_*}{f} = \lambda. \quad (11)$$

A significant new result is obtained in conditions of strong hydrostatic instability, i.e., at large negative μ 's. At heights roughly estimated as $z > 0.1|L|$, free convection conditions develop, and the eddy viscosity is expressed by the well-known formula

$$k = \frac{3}{C_u} \kappa^{4/3} (\beta Q)^{1/3} z^{4/3}, \quad (12)$$

where $C_u \approx 1.2$ is a non-dimensional constant (see Zilitinkevich, 1970). Finding K once more through substituting h_τ for z in the formula for k and using (10), we conclude that

$$h_\tau \sim \kappa^2 \left(\frac{3}{C_u}\right)^{3/2} (\beta Q)^{1/2} f^{-3/2} = \left(\frac{3}{C_u}\right)^{3/2} \lambda \sqrt{-\mu}. \quad (13)$$

Thus, *with increasing instability, the Ekman layer thickness increases as $\sqrt{-\mu}$* . The numerical factor in the right-hand part of (13) should not be seriously considered, for, as pointed out above, the proportionality factor is actually determined in this case by the selection of a tolerable error δ_τ in (7).

It should be noted that a formula equivalent to (13) is obtained directly from dimensional analysis, assuming that h_τ is determined by the Q , β and f parameters, i.e., that it is independent of u_* .

We should now discuss stable stratification when μ acquires large positive values. As is known, the flux Richardson number

$$R_f = -\frac{\beta Q \varrho}{\tau_x \frac{\partial u}{\partial z} + \tau_y \frac{\partial v}{\partial z}} = -\frac{\beta Q \varrho^2}{\tau_x^2 + \tau_y^2} k \quad (14)$$

* The height h_u is not so simple to express and depends on ν , f , and $\sqrt{(U_g^2 + V_g^2)}$.

cannot exceed a certain critical value R which is a non-dimensional universal constant (the experimental data yield a value of $R \approx 10$; see Zilitinkevich, 1970). Hence, the value of the eddy viscosity k during stable stratification is constrained according to the inequality

$$k \leq -\frac{\tau_x^2 + \tau_y^2}{\rho^2} \frac{R}{\beta Q} \leq -\frac{Ru_*^4}{\beta Q}. \quad (15)$$

(In the flow under consideration, the momentum flux decreases with height, therefore $(\tau_x^2 + \tau_y^2)/\rho^2 \leq u_*^4$.) Using the right-hand part of (15) to estimate the effective value of the eddy viscosity, i.e., assuming $K = -Ru_*^4/\beta Q$ and taking account of (10), we find

$$h_\tau \sim \sqrt{-\frac{Ru_*^4}{\beta Q f}} = \lambda \sqrt{\frac{R}{\mu}}. \quad (16)$$

Thus, with increasing hydrostatic stability, the Ekman layer height decreases in inverse proportion to $\sqrt{\mu}$.

As to the height of the Ekman boundary layer h_u found by the velocity defect law, general considerations permit inferences concerning its behaviour to be made only at large instabilities, i.e., at large negative μ values. In this instance major variations of velocity with height are confined to the ground layer, and, in accordance with the Monin-Obukhov similarity theory, the expression for the velocity defect may be approximately written as:

$$\sqrt{(u - U_g)^2 + (v - V_g)^2}/u_* \approx \frac{C_u}{\kappa^{4/3}} \frac{u_*}{(\beta Q z)^{1/3}}. \quad (17)$$

Taking into account the first formula (7), we find

$$h_u \approx \frac{(C_u u_*)^3}{\kappa^4 \delta_u^3 \beta Q} = -\left(\frac{C_u}{\kappa \delta_u}\right)^3 \frac{\lambda}{\mu}. \quad (18)$$

Therefore, the behaviour of the φ_τ and φ_u functions proves to be strikingly different in unstable stratification.

The conclusions provided by the formulae (11), (13) and (16) concerning the nature of the dependence of h_τ/λ on μ and the order of magnitude of h_τ agree qualitatively with observational data in the atmospheric boundary layer. However, because experimental data have not been presented in a way suitable for testing similarity formula of the type given by eq. (8), no information is available yet for a quantitative test of the results of the present paper.

References

- Hanna, S. R.: 1969, 'The Thickness of the Planetary Boundary Layer', *Atmos. Envir.* **3**, 519–536.
 Kazanski, A. B. and Monin, A. S.: 1960, *On Turbulent Conditions above the Atmospheric Surface Layer*, *Izvestia Akad. Nauk S.S.S.R., Geophys. Series*, No. 1, 165–168.
 Monin, A. S. and Obukhov, A. M.: 1954, *Basic Laws of Turbulent Mixing in the Ground Layer of the Atmosphere*, *Trudy Geofiz. Inst. Akad. Nauk S.S.S.R.*, No. 24 (151), 163–187.
 Monin, A. S. and Yaglom, A. M.: 1971, *Statistical Fluid Mechanics*, Vol. 1, MIT Press, Cambridge Mass., 769 pp.
 Zilitinkevich, S. S.: 1970, *Dynamics of the Atmospheric Boundary Layer*, Gidrometeoizdat, Leningrad.